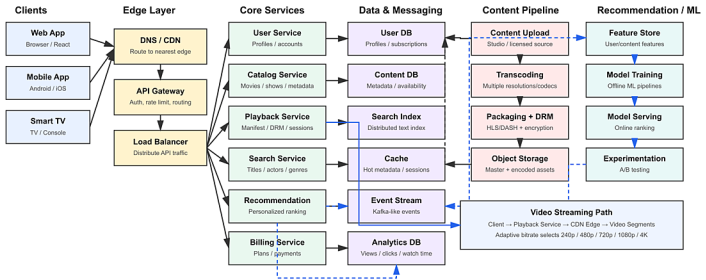


Netflix — High-Level System Design

Scalable video streaming, content delivery, recommendations and playback architecture



Key Design Decisions

Scalability

- Stateless API services scale horizontally
- CDN handles massive video traffic
- Cache hot metadata and sessions
- Partition large datasets by user/content
- Async event processing for analytics
- Multi-region deployment for resilience

Reliability

- Replicated databases
- Health checks + automatic failover
- Retries with exponential backoff
- Circuit breakers between services
- Idempotent payment operations
- Monitoring, logging and tracing

Performance

- CDN serves video close to users
- Adaptive bitrate avoids buffering
- Prefetch next video segments
- Search index for fast discovery
- Precompute recommendation candidates
- Keep critical APIs low-latency

Security

- TLS for network communication
- OAuth/session authentication
- DRM-protected video playback
- Tokenized playback URLs
- Rate limiting and abuse detection
- Encrypt sensitive data at rest

High-level educational architecture — components are logical and can be implemented with different technologies.